



Comsats University, Islamabad, Attock
Campus

**Department Of
Computer Science**

ASSIGNMENT NO : 4

Subject: MAD (Theory)

Program: BSSE-VI

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Reg N0: SP22-BSE-051

Date: 24 December,2024

CODE:

```
import React, { useEffect } from 'react';
import {
  View,
  Text,
  TextInput,
  TouchableOpacity,
  Alert,
  StyleSheet,
  ScrollView,
  Image,
} from 'react-native';
import { NavigationContainer } from '@react-navigation/native';
import { createStackNavigator } from '@react-navigation/stack';
import { Provider, useDispatch, useSelector } from 'react-redux';
import { configureStore, createSlice } from '@reduxjs/toolkit';
import AsyncStorage from '@react-native-async-storage/async-storage';
import * as Location from 'expo-location';
const FIREBASE_API_ENDPOINT = 'https://myproject-37990-default-rtdb.firebaseio.com/';
const Stack = createStackNavigator();
const userSlice = createSlice({
  name: 'user',
  initialState: {
    name: '',
    email: '',
    phoneNumber: '',
    password: '',
    isAuthenticated: false,
    location: null,
  },
  reducers: {
    setName: (state, action) => {
      state.name = action.payload;
    },
    setEmail: (state, action) => {
      state.email = action.payload;
    },
    setPhoneNumber: (state, action) => {
      state.phoneNumber = action.payload;
    },
    setPassword: (state, action) => {
      state.password = action.payload;
    },
    setAuthentication: (state, action) => {
      state.isAuthenticated = action.payload;
    },
    setLocation: (state, action) => {
      state.location = action.payload;
    },
  },
});
```

```

},
});
const { setName, setEmail, setPhoneNumber, setPassword, setAuthentication, setLocation } =
userSlice.actions;
const store = configureStore({
  reducer: {
    user: userSlice.reducer,
  },
});
// Signup Screen
const SignupScreen = ({ navigation }) => {
  const { name, email, phoneNumber, password } = useSelector((state) => state.user);
  const dispatch = useDispatch();
  const validateEmail = (email) => /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email);
  const validatePhoneNumber = (number) => /^[0-9]{10}$/.test(number);
  const handleSignup = async () => {
    if (!name || !email || !phoneNumber || !password) {
      Alert.alert('Error', 'Please fill in all fields');
    } else if (!validateEmail(email)) {
      Alert.alert('Error', 'Invalid email syntax');
    } else if (!validatePhoneNumber(phoneNumber)) {
      Alert.alert('Error', 'Invalid phone number. It should be 10 digits.');
```

} else if (password.length < 6) {
 Alert.alert('Error', 'Password must be at least 6 characters');
 } else {
 try {
 const response = await fetch(`\${FIREBASE\_API\_ENDPOINT}/users.json`, {
 method: 'POST',
 body: JSON.stringify({ name, email, phoneNumber, password }),
 headers: { 'Content-Type': 'application/json' },
 });
 if (response.ok) {
 dispatch(setAuthentication(true));
 Alert.alert('Success', 'Signup successful!');
 navigation.navigate('Login');
 } else {
 Alert.alert('Error', 'Failed to insert data.');

}
 } catch (error) {
 Alert.alert('Error', 'Something went wrong.');

}
 }
 };
 return (
 <View style={styles.container}>
 <Text style={styles.title}>Signup</Text>
 <TextInput
 style={styles.input}
 placeholder="Name"
 value={name}
 onChangeText={(text) => dispatch(setName(text))}>
 </Text>
 <TextInput
 style={styles.input}

```

placeholder="Email"
keyboardType="email-address"
value={email}
onChangeText={(text) => dispatch(setEmail(text))}
/>
<TextInput
style={styles.input}
placeholder="Phone Number"
keyboardType="phone-pad"
value={phoneNumber}
onChangeText={(text) => dispatch(setPhoneNumber(text))}
/>
<TextInput
style={styles.input}
placeholder="Password"
secureTextEntry
value={password}
onChangeText={(text) => dispatch(setPassword(text))}
/>
<TouchableOpacity style={styles.button} onPress={handleSignup}>
<Text style={styles.buttonText}>Signup</Text>
</TouchableOpacity>
</View>
);
};
// Login Screen
const LoginScreen = ({ navigation }) => {
const { email, password } = useSelector((state) => state.user);
const dispatch = useDispatch();
const fetchLocation = async () => {
const { status } = await Location.requestForegroundPermissionsAsync();
if (status !== 'granted') {
Alert.alert('Error', 'Permission to access location was denied.');
```

```

latitude: location.coords.latitude,
longitude: location.coords.longitude,
};
await AsyncStorage.setItem('userLocation', JSON.stringify(locationData));
dispatch(setLocation(locationData));
}
dispatch(setAuthentication(true));
Alert.alert('Success', 'Login successful!');
navigation.navigate('Profile');
} else {
Alert.alert('Error', 'Invalid email or password');
}
} else {
Alert.alert('Error', 'No users found. ');
}
} catch (error) {
Alert.alert('Error', 'Something went wrong. ');
}
}
};
return (
<View style={styles.container}>
<Text style={styles.title}>Login</Text>
<TextInput
style={styles.input}
placeholder="Email"
keyboardType="email-address"
value={email}
onChangeText={(text) => dispatch(setEmail(text))}
/>
<TextInput
style={styles.input}
placeholder="Password"
secureTextEntry
value={password}
onChangeText={(text) => dispatch(setPassword(text))}
/>
<TouchableOpacity style={styles.button} onPress={handleLogin}>
<Text style={styles.buttonText}>Login</Text>
</TouchableOpacity>
</View>
);
};
// Profile Screen
const ProfileScreen = () => {
  const { name, email, phoneNumber, location } = useSelector((state) => state.user);
  const [storedLocation, setStoredLocation] = React.useState(null);

  useEffect(() => {
    const fetchStoredLocation = async () => {
      try {
        const locationData = await AsyncStorage.getItem('userLocation');
        if (locationData) {
          setStoredLocation(JSON.parse(locationData));
        }
      } catch (error) {
        console.log('Error fetching stored location', error);
      }
    };
    fetchStoredLocation();
  }, []);

  const handleLogout = () => {
    dispatch(setAuthentication(false));
    navigation.navigate('Login');
  };

  return (
    <View style={styles.container}>
      <Text style={styles.title}>Profile</Text>
      <Text style={styles.name}>Name: {name}</Text>
      <Text style={styles.email}>Email: {email}</Text>
      <Text style={styles.phone}>Phone: {phoneNumber}</Text>
      <Text style={styles.location}>Location: {location}</Text>
      <Text style={styles.logout}>Logout</Text>
    </View>
  );
};

```

```

    }
  } catch (error) {
    console.error('Failed to load location from storage', error);
  }
};

fetchStoredLocation();
}, []);

return (
  <ScrollView contentContainerStyle={styles.scrollContent}>
    <View style={styles.header}>
      <Text style={styles.title}>My Profile</Text>
    </View>
    <View style={styles.profileSection}>
      <Image
        source={{ uri: 'https://randomuser.me/api/portraits/men/1.jpg' }}
        style={styles.profileImage}
      />
      <Text style={styles.profileName}>{name || 'John Doe'}</Text>
      <Text style={styles.profileDetails}>{email}</Text>
      <Text style={styles.profileDetails}>{phoneNumber}</Text>
      {storedLocation && (
        <View style={styles.locationContainer}>
          <Text style={styles.locationText}>Location:</Text>
          <Text style={styles.locationCoordinates}>
            Latitude: {storedLocation.latitude.toFixed(4)}
          </Text>
          <Text style={styles.locationCoordinates}>
            Longitude: {storedLocation.longitude.toFixed(4)}
          </Text>
        </View>
      )}
    </View>
    <View style={styles.ordersSection}>
      <View style={styles.orderHeader}>
        <Text style={styles.sectionTitle}>Recent Orders</Text>
        <TouchableOpacity>
          <Text style={styles.seeAllText}>See All</Text>
        </TouchableOpacity>
      </View>
      <View style={styles.orderItem}>
        <Image
          source={{
            uri:
'https://upload.wikimedia.org/wikipedia/commons/4/45/A_small_cup_of_coffee.JPG',
          }}
          style={styles.orderImage}
        />
        <View style={styles.orderDetails}>
          <Text style={styles.orderTitle}>Coffee, Sandwich, Muffin</Text>
          <Text style={styles.orderDate}>22 Thu 10.00</Text>
        </View>
        <Text style={styles.orderPrice}>$12.50</Text>
      </View>
    </View>
  </ScrollView>
);

```

```

    </View>
    <View style={styles.orderItem}>
      <Image
        source={{
          uri: 'https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcTjQFf31CNn3kmzrUBVZpxYUQBZ1rBzOv1jKA&s',
        }}
        style={styles.orderImage}
      />
      <View style={styles.orderDetails}>
        <Text style={styles.orderTitle}>Donut, Pancake, Juice</Text>
        <Text style={styles.orderDate}>15 Fri 18.00</Text>
      </View>
      <Text style={styles.orderPrice}>$20.30</Text>
    </View>
    <View style={styles.orderItem}>
      <Image
        source={{
          uri: 'https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcQMrD61WPfd_39zCAKmo0DatnyZf9VZXiNqvA&s',
        }}
        style={styles.orderImage}
      />
      <View style={styles.orderDetails}>
        <Text style={styles.orderTitle}>Beef Burger, Fries, Coke</Text>
        <Text style={styles.orderDate}>11 Mon 10.00</Text>
      </View>
      <Text style={styles.orderPrice}>$25.75</Text>
    </View>
  </View>
</ScrollView>
);
};

// App Navigation
const App = () => {
  return (
    <Provider store={store}>
      <NavigationContainer>
        <Stack.Navigator initialRouteName="Signup">
          <Stack.Screen name="Signup" component={SignupScreen} />
          <Stack.Screen name="Login" component={LoginScreen} />
          <Stack.Screen name="Profile" component={ProfileScreen} />
        </Stack.Navigator>
      </NavigationContainer>
    </Provider>
  );
};

// Styles
const styles = StyleSheet.create({
  container: {
    flex: 1,
    backgroundColor: '#F7F7F7',
    padding: 20,
  },
});

```

```
},
scrollContent: {
  paddingBottom: 20,
},
header: {
  alignItems: 'center',
  marginBottom: 20,
},
title: {
  fontSize: 24,
  fontWeight: 'bold',
  marginBottom: 20,
  textAlign: 'center',
  color: '#333',
},
input: {
  borderWidth: 1,
  borderColor: '#ccc',
  padding: 10,
  marginBottom: 15,
  borderRadius: 5,
  backgroundColor: '#FFF',
  fontSize: 16,
  color: '#333',
},
button: {
  backgroundColor: '#007BFF',
  padding: 15,
  borderRadius: 5,
  alignItems: 'center',
  marginTop: 10,
},
buttonText: {
  color: '#FFF',
  fontSize: 16,
  fontWeight: 'bold',
},
profileSection: {
  backgroundColor: '#FFF',
  padding: 20,
  borderRadius: 10,
  alignItems: 'center',
  marginBottom: 20,
  shadowColor: '#000',
  shadowOffset: { width: 0, height: 2 },
  shadowOpacity: 0.1,
  shadowRadius: 5,
  elevation: 2,
},
profileImage: {
  width: 80,
  height: 80,
  borderRadius: 40,
  marginBottom: 10,
```



```
},
profileName: {
  fontSize: 18,
  fontWeight: 'bold',
  marginBottom: 5,
  color: '#333',
},
profileDetails: {
  fontSize: 16,
  color: '#666',
  marginBottom: 5,
},
ordersSection: {
  marginTop: 20,
},
sectionTitle: {
  fontSize: 20,
  fontWeight: 'bold',
  marginBottom: 10,
  color: '#333',
},
orderItem: {
  flexDirection: 'row',
  backgroundColor: '#FFF',
  padding: 15,
  borderRadius: 5,
  marginBottom: 10,
  alignItems: 'center',
  shadowColor: '#000',
  shadowOffset: { width: 0, height: 2 },
  shadowOpacity: 0.1,
  shadowRadius: 5,
  elevation: 2,
},
orderImage: {
  width: 60,
  height: 60,
  borderRadius: 5,
  marginRight: 15,
},
orderDetails: {
  flex: 1,
},
orderTitle: {
  fontSize: 16,
  fontWeight: 'bold',
  color: '#333',
  marginBottom: 5,
},
orderDate: {
  fontSize: 12,
  color: '#666',
},
orderPrice: {
```

```
fontSize: 16,
fontWeight: 'bold',
color: '#333',
marginLeft: 10,
},
// Recent Orders Header
orderHeader: {
flexDirection: 'row',
justifyContent: 'space-between',
alignItems: 'center',
marginBottom: 10,
},
seeAllText: {
fontSize: 14,
color: '#007BFF',
},
locationContainer: {
marginTop: 15,
padding: 10,
borderRadius: 8,
backgroundColor: '#e0f7fa',
alignItems: 'center',
},
locationText: {
fontSize: 18,
fontWeight: 'bold',
color: '#00796b',
marginBottom: 5,
},
locationCoordinates: {
fontSize: 16,
color: '#004d40',
},
});
export default App;
```

OUTPUT:

Signup

Signup

Signup

Login

Login

Login

Profile



MARYUM

MARYUM2@gmail.com

3569852547

Location:

Latitude: 33.8613

Longitude: 72.4423

Recent Orders

[See All](#)



Coffee, Sandwich, Muffin

22 Thu 10.00

\$12.50



Donut, Pancake, Juice

15 Fri 18.00

\$20.30



Beef Burger, Fries, Coke

11 Mon 10.00

\$25.75

LOCATION:

snack-web-player.s3.us-west-1.amazonaws.com wants to

Know your location

Allow

Block

FIREBASE:

-0EsHohUwa9mZ3up0E0P

email: "MARYUM2@gmail.com"

name: "Maryum"

password: "help21234"

phoneNumber: "3569852547"

Database location: United States (us-central1)